

NABILA NEWAZ

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PROFESSIONAL SUMMARY

Computer Science & Engineering graduate from IUT who builds full-stack and AI-powered applications: Node.js/Express, FastAPI and Django back ends; React/Next.js and Flutter front ends; and PostgreSQL, Supabase, MongoDB and Firebase data layers. Hands-on with authentication and row-level security, concurrency control and integration testing, RAG pipelines and LLM features, and undergraduate research on LLM reasoning. Experienced with Docker, Git-based workflows and Vercel deployments, distributed cross-time-zone teams, and AI-assisted development.

EDUCATION

Bachelor of Science in Computer Science and Engineering CGPA 3.71

Islamic University of Technology (IUT), Gazipur

Relevant coursework: Data Structures, Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Algorithm Engineering

Higher Secondary Certificate (HSC) · Rajuk Uttara Model College, Dhaka GPA 5.00

Secondary School Certificate (SSC) · Rajuk Uttara Model College, Dhaka GPA 5.00

EXPERIENCE

AI / ML Intern EATL

Personalized Study Planner

FastAPI · Python · React · TypeScript · Tailwind CSS · FAISS · Gemini · Llama

- Built a web application that analyzes students' past performance and upcoming exam dates to generate exam-aligned study plans, with a topic-level syllabus breakdown, estimated study hours and linked resources.
- Designed the FastAPI backend and a retrieval-augmented generation (RAG) pipeline: indexed books and notes into a FAISS vector store and generated source-grounded plans with Gemini and Llama.
- Tuned prompts and workflows against defined test scenarios; released to real students with no critical post-launch defects.
- Reported implementation and validation results to stakeholders in non-technical terms, reaching sign-off with no unresolved critical issues.

Mobile App Development Intern Excelebrate (remote, international team)

- Built and self-tested cross-platform Flutter features across structured sprint cycles, delivering on schedule to a team spread across time zones.
- Found and fixed integration bugs before merge, reducing the issues that reached review.

Data Visualization Associate Intern Excelebrate (remote, international team)

- Built Power BI dashboards and validated every figure against the source datasets, leaving no unresolved discrepancies.

KEY PROJECTS

Dhaka Tesla Pool: Ride-Pooling Platform

Next.js · Express 5 · TypeScript · PostgreSQL · Prisma · Zod · Vitest · Supertest · Docker · Vercel · Neon

- Developed a full-stack ride-pooling platform where passengers heading the same way share a 3-seat vehicle, each paying an individual fare calculated in integer paisa.
- Prevented overbooking under concurrent bookings with an atomic conditional `UPDATE`, row locks taken in a fixed order and a database `CHECK` constraint; proved it with race tests (8 riders for 2 seats) and mutation tests.
- Modeled trips and rides as two linked state machines with an append-only audit trail; 125 unit and integration tests, the integration tests running against a real PostgreSQL database.
- Designed a Bangla-first roadside interface for drivers (per-passenger drop-off, early drop-off fares, breakdown handling) and 30 pickup landmarks verified on OpenStreetMap, priced per 100 m.
- Shipped with one-command Docker Compose (migrations, seed data, health checks) and deployed on Vercel with Neon PostgreSQL through a feature → pre-release → release branching workflow.

PaperVault: AI-Powered Research Paper Vault

React · TypeScript · Vite · Tailwind CSS · shadcn/ui · TanStack Query · Supabase (PostgreSQL, Auth, Storage, Edge Functions) · Groq · Vercel

- Built and deployed a web app for Computer Science students to find, organize and study research papers: search Semantic Scholar and arXiv, upload PDFs, sort papers into collections and track reading progress (To Read, Reading, Done).
- Generated summaries, key points and flashcards with an LLM (Groq) through a Supabase Edge Function, so the API key never reaches the browser; linked papers to their GitHub code via Papers With Code and suggested related papers.

- Protected user data with Supabase Auth (JWT) and PostgreSQL Row-Level Security: users can read, add, change and delete only their own papers, and upload or delete only their own PDFs.
- Kept the app usable when Supabase is unreachable with an offline fallback to localStorage and IndexedDB; every push to main deploys automatically to Vercel.

BizAssist: AI-Powered Business Strategy Platform

Node.js · Express · TypeScript · Next.js · Firebase · Gemini · Cloudinary

- Platform that helps entrepreneurs turn business ideas into investor-ready pitch decks.
- Designed a multi-stage evaluation pipeline that scores business ideas on market potential, feasibility, scalability and profitability by chaining Gemini calls through an Express backend.
- Integrated the Next.js/TypeScript frontend with the REST API and checked data consistency at each stage.

Shondhan: Augmented-Reality Property Finding App

Flutter · Dart · Firebase / Firestore · Google Maps API · ARCore · Unity · Flask · scikit-learn

- Mobile app that makes property search interactive: real-time AR visualization, map-based discovery, natural-language search, and crime-score and price prediction for each listing.
- Built the property flows in Flutter: the add-new-property and dynamic property screens, and a property details screen that shows the predicted crime score and price.
- Added a launcher that opens a Unity-based 3D architectural visualization, so buyers can walk through a property's interior.

TechHive: Full-Stack E-Commerce Platform

React · Tailwind CSS · Node.js · Express · MongoDB · JWT

- Developed the Express.js routes and controllers behind admin order management, exposing REST endpoints consumed by the React admin panel.
- Designed and built the admin panel UI (layout, header, sidebar, orders view) in React and Tailwind CSS, integrating JWT-authenticated API calls.
- Implemented product reviews and star ratings end to end, from the MongoDB data model through the Express API to the React UI.

LyRicA: Music Streaming Platform

Python · Django · PostgreSQL · HTML · CSS · JavaScript · pCloud

- Web app for streaming music, with sign-up and login, favorites, top songs ranked by stream counts, and browsing by artist, album and genre.
- Developed user profiles (view and edit), the artist, album and genre listings, and artist following.
- Implemented the song-request workflow end to end: users request songs; admins review requests, upload the tracks (stored in pCloud) and edit or delete songs; artists upload their own music.

RESEARCH EXPERIENCE

Undergraduate Thesis: Latent-Space Steering in Compressed & Continuous Chain-of-Thought Reasoning

Islamic University of Technology (IUT)

Python · PyTorch · Hugging Face · LoRA · GSM8K

- Recovered accuracy lost to reasoning compression on the full GSM8K test set (1,319 problems), from **69.0% to 72.0%**, with multi-layer activation-steering directions; validated against random-noise and shuffled-label controls.
- Ran a four-phase research pipeline across three model backbones and two reasoning architectures, with strictly isolated train, steering, validation and locked test splits (no data leakage).

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C++, C, Dart, SQL, HTML, CSS

Backend & APIs: Node.js, Express, FastAPI, Flask, Django, REST API design, JWT authentication, Prisma, Zod

Frontend & Mobile: React, Next.js, Tailwind CSS, Flutter

Databases: PostgreSQL, MySQL, MongoDB, Firebase / Firestore, Supabase

Testing & DevOps: Vitest, Supertest, Docker / Docker Compose, Git & GitHub, Linux (Ubuntu), Vercel

AI / ML: PyTorch, Hugging Face, LoRA, activation steering, RAG pipelines, FAISS, LangChain, Gemini, Llama

Tools: VS Code, Postman, Power BI, Figma, Jupyter Notebook, Excel

ACHIEVEMENTS & ACTIVITIES

- 4th Place, CalorieQuest Machine Learning Competition
- Top 12, KUET BitFest Datathon
- Participant, SOLVIO AI Hackathon (Sheba Platform)
- Participant, UIU CSE Fest 2025 Project Show
- Member, IUT Computer Society
- Mentee, SheSTEM Mentorship Program